Power law, Paul's Law or critical power?

Predicting erg performance with one- and two-parameter models

Verified Concept2 RowErg rankings, season 2026

Generated 2026-11-02

1 Summary

The power law validated in the companion analysis, $t = k\,d^{1.064}$, has one free parameter per athlete (the level k). Paul's Law is its best-known rival: pace rises by a fixed number of seconds, 5 s per 500 m, for every doubling of distance. We compare both with each other and with the two-parameter critical-power (CP) model, using the same 3,993 athlete-seasons as the companion analysis. Each model is fitted to an athlete's other efforts and asked to predict the time for the distance of a held-out effort (leave-one-event-out); 17,230 predictions are common to the nine main models.

- **Overall accuracy.** The lowest median absolute error of the predicted time is 2.14%, for PP2 (free constant). PL1 has 2.25% (rank 6 of 9), Paul's Law with its published constant PP1 has 2.19% (rank 2), and the best CP variant 3.02%.
- **Paul's Law against the power law.** With the published constant, Paul's Law is more accurate than the power law with exponent 1.064 (2.19% against 2.25%; difference −0.07 percentage points [−0.10, −0.04]). With the constant (or exponent) calibrated on other athletes by sex and age group, Paul's Law is not clearly different from the power law (2.22% against 2.23%; difference 0.00 percentage points [−0.03, 0.02]). With a free constant or exponent per athlete (two parameters), Paul's Law is more accurate than the power law (2.14% against 2.20%; difference −0.05 percentage points [−0.08, −0.03]).
- **The constant.** The constant that best fits these athletes is 5.48 s per 500 m per doubling [5.38, 5.57] against the published 5 s. It stays close to fixed across paces, as Paul's Law assumes: a constant fits the pace bins with weighted SSE 11, a constant proportional to pace (a power law, exponent 1.070) with 149.
- **From the 2 km result alone,** Paul's Law is more accurate than the power law (2.29% against 2.47%; difference −0.18 percentage points [−0.22, −0.14]).
- **Critical power.** The classical CP fit is less accurate than PL1 (3.13% against 2.25%; difference 0.88 percentage points [0.80, 0.98]); fitted on log time like the other models, CP is less accurate than PL1 (3.02% against 2.25%; difference 0.76 percentage points [0.68, 0.84]).

2 Paul's Law as a power law

Paul's Law in its usual form predicts the time t_1 over distance d_1 from a known effort t_0 over d_0:

$$t_1 = d_1\left(\frac{t_0}{d_0} + \frac{1}{100}\log_2\left(\frac{d_1}{d_0}\right)\right).$$

Written in pace $p = \frac{t}{d}$ (seconds per metre) it adds $1/100$ s per metre for each doubling of distance, i.e. 5 s per 500 m. The power law multiplies the pace instead: $p_1 = p_0(d_1/d_0)^{\beta-1}$. For a distance ratio $r = d_1/d_0$ close to 1, $r^{\beta-1} \approx 1 + (\beta-1)\ln r$, so the two laws agree when

$$\beta = 1 + \frac{c}{p_0 \ln 2},$$

with c Paul's constant (5 s per 500 m) and p_0 the athlete's pace. Paul's Law is therefore a power law whose exponent depends on the athlete's pace: the slower the athlete, the smaller the exponent. An exponent of 1.064 corresponds to a pace of 112.7 s per 500 m, which is a 2 km time of 7:30.8.

The equivalence is local. Paul's pace increase is linear in $\ln d$, the power law's is exponential, so over a finite range the equivalent exponent is a little smaller than the local one. Table 1 gives both.

Table 1: Power-law exponent equivalent to Paul's Law (5 s per 500 m per doubling) for an athlete with a given 2 km time: local exponent and exponent that gives the same prediction over a given distance ratio.

2 km time	Pace (s/500 m)	Local exponent	2 km → 6 km	2 km → 10 km	2 km → marathon
6:00	90.0	1.080	1.077	1.075	1.072
6:30	97.5	1.074	1.071	1.070	1.067
7:00	105.0	1.069	1.066	1.065	1.062
7:30	112.5	1.064	1.062	1.061	1.059
8:00	120.0	1.060	1.058	1.057	1.055
8:30	127.5	1.057	1.055	1.054	1.052
9:00	135.0	1.053	1.052	1.051	1.050
9:30	142.5	1.051	1.049	1.049	1.047
10:00	150.0	1.048	1.047	1.046	1.045

Two consequences can be checked against the data. First, Paul's Law predicts that exponents are smaller for slower athletes. Table 2 compares the exponent Paul's Law implies for the median 2 km time of the analysed men and women with the pooled exponents estimated in the companion analysis.

Table 2: Exponent implied by Paul's Law for the median analysed athlete of each sex, against the observed pooled exponent.

Sex	Median 2 km time (s)	Pace (s/500 m)	Paul-implied exponent	Observed pooled exponent	Observed – implied
Women	531.1	132.8	1.054	1.060	0.006
Men	449.1	112.3	1.064	1.069	0.004

Second, Paul's Law makes the exponent fall with distance: the local exponent changes by $-(\beta-1)^2$ per unit of $\ln d$ even without any extra parameter, a built-in concavity of the log-log curve that a constant-exponent power law lacks. The companion analysis found such a concavity in the data.

3 Models

All models describe one athlete in one season and are fitted separately for each athlete. Average power on the Concept2 RowErg is $P = 2.8\,(d/t)^3$; the constant 2.8 does not affect any prediction because the CP parameters rescale with it.

Table 3: Models compared.

Model	Free parameters	Prediction	Fitting
PL1	1 (level)	$t = k\,d^{1.064}$	geometric mean of the training efforts
PL1 calibrated	1 (level)	$t = k\,d^{\beta_g}$	β_g: pooled exponent of the athlete's sex and age group from other athletes only (10-fold cross-fitting)
PL1 from 2 km	1 (2 km time)	$t = t_{2000}(d/2000)^{1.064}$	the 2 km result alone
PL2	2	$\ln t = a + b\ln(d/2000)$	least squares on log time
PP1	1 (level)	$p = p_{2000} + c\log_2(d/2000)$, c = 5 s per 500 m	level by least squares on log time
PP calibrated	1 (level)	as PP1 with c_g	c_g: pooled within-athlete regression of pace on $\log_2 d$ for the athlete's sex and age group, from other athletes only
PP from 2 km	1 (2 km time)	$t_1 = d_1\big(t_0/d_0 + \frac{1}{100}\log_2(d_1/d_0)\big)$	the 2 km result alone
PP2	2 (level, \$c\$)	as PP1	least squares on log time
CP, classical	2 (CP, W′)	$P = \mathrm{CP} + W'/t$, i.e. CP $t^3 + W't^2 = 2.8d^3$	regression of work $W = Pt$ on t, CP and W′ ≥ 0
CP, log-time fit	2 (CP, W′)	as above	least squares on log time
CP, 2–20 min	2 (CP, W′)	as above	classical fit on training efforts of 2–20 minutes (at least two needed)
Critical speed	2 (CS, D′)	$d = D' + \mathrm{CS}\,t$	regression of d on t, D′ ≥ 0

The CP cubic is solved numerically for the time at the held-out distance, so every model answers the same question: *how long would this athlete take for this distance?* A predicted pace below 0.01 s per metre (an impossible 5 s per 500 m) is floored, so that a failed Paul fit counts as a large error rather than being dropped. The calibrated variants use constants estimated on other athletes only, so they cost the athlete's own held-out result nothing. Where a one-parameter power law and a one-parameter Paul's Law are compared, they are fitted the same way: level by least squares on log time, the shape constant fixed.

4 Evaluation design

Data. The same athlete-seasons as the companion analysis: verified individual RowErg results from season 2026 with at least 3 events spanning a distance ratio of at least 3 (3,993 athlete-seasons, 17,446 results).

Leave-one-event-out. Each event is held out in turn, every model is fitted to the athlete's remaining events and predicts the held-out result, giving 17,446 predictions. Two-parameter models therefore have at least two training efforts. For a timed event (4, 30 or 60 minutes) the held-out result is the distance rowed in a fixed time; the model predicts the time for that distance and is compared with the fixed time.

Error and uncertainty. The error is the predicted time divided by the observed time minus one, in percent. We report the median absolute error, its 90th percentile, the median signed error (bias) and the share of predictions within 1% and 2%. Intervals come from a cluster bootstrap (1,000 resamples) that resamples athletes. Model comparisons are paired: the difference of median absolute errors is bootstrapped on the same predictions.

Interpolation or extrapolation. A held-out result is an interpolation if its duration lies between the shortest and longest training efforts. 55% of predictions are interpolations, 22% extrapolate to shorter and 23% to longer efforts.

Prediction sets. The nine main models are compared on the 17,230 predictions for which all are defined. The models that need a 2 km result are compared on the 9,803 predictions where the athlete has a 2 km result among the training efforts, and CP restricted to 2–20 minute efforts on the 8,901 predictions where it exists as well.

5 Results

5.1 Overall accuracy

Table 4: Out-of-sample accuracy on 17,230 predictions (median absolute error of the predicted time; difference in percentage points with 95% bootstrap interval; "wins" is the share of predictions, excluding ties, where the model is closer than PL1).

Model	Median error	90th pct	Bias	Within 1%	Within 2%	Difference vs PL1	Wins vs PL1
PP2 (free constant)	2.14%	8.7%	0.15%	27%	48%	−0.11 [−0.17, −0.05]	51%
PP1 (Paul, 5 s per 500 m)	2.19%	7.6%	0.06%	25%	47%	−0.07 [−0.10, −0.04]	53%
PL2 (free exponent)	2.20%	8.8%	0.16%	26%	47%	−0.06 [−0.12, 0.00]	50%
PP (calibrated constant)	2.22%	7.3%	0.16%	25%	46%	−0.03 [−0.07, 0.00]	52%
PL1 (calibrated exponent)	2.23%	7.4%	0.12%	25%	46%	−0.03 [−0.06, 0.00]	53%
PL1 (exponent 1.064)	2.25%	7.6%	0.09%	24%	45%	–	–
CP (log-time fit)	3.02%	11.5%	−0.29%	19%	36%	0.76 [0.68, 0.84]	39%
CP (classical fit)	3.13%	13.8%	−0.48%	19%	35%	0.88 [0.80, 0.98]	38%
Critical speed	3.51%	20.7%	−0.60%	17%	33%	1.25 [1.15, 1.34]	34%

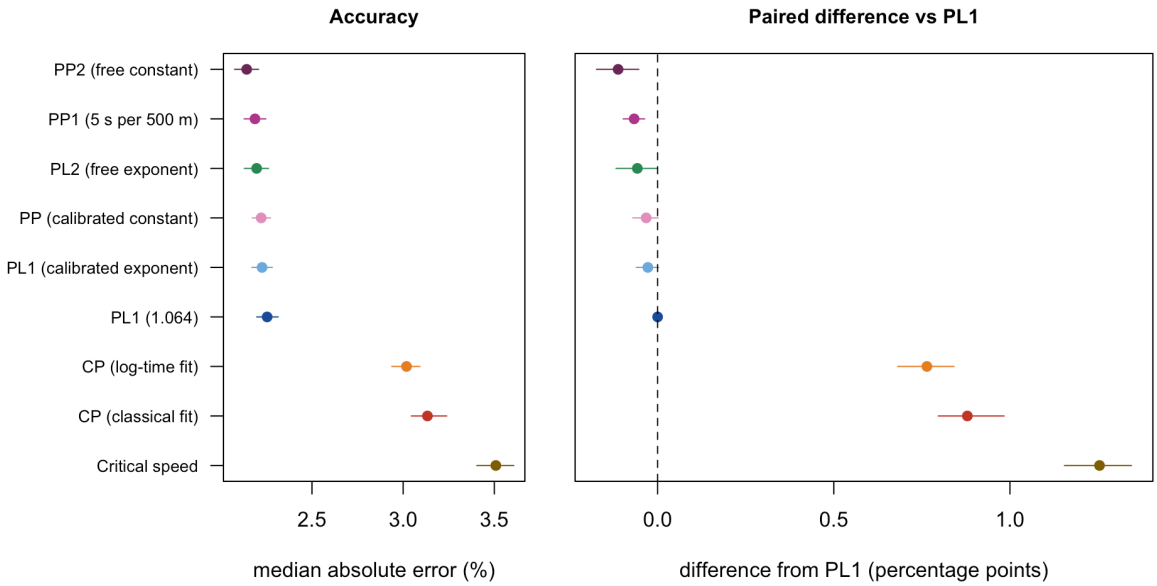


Figure 1: Median absolute error with 95% intervals (left) and paired difference from PL1 (right).

Figure 2 shows what the fitted curves look like for four athlete-seasons.

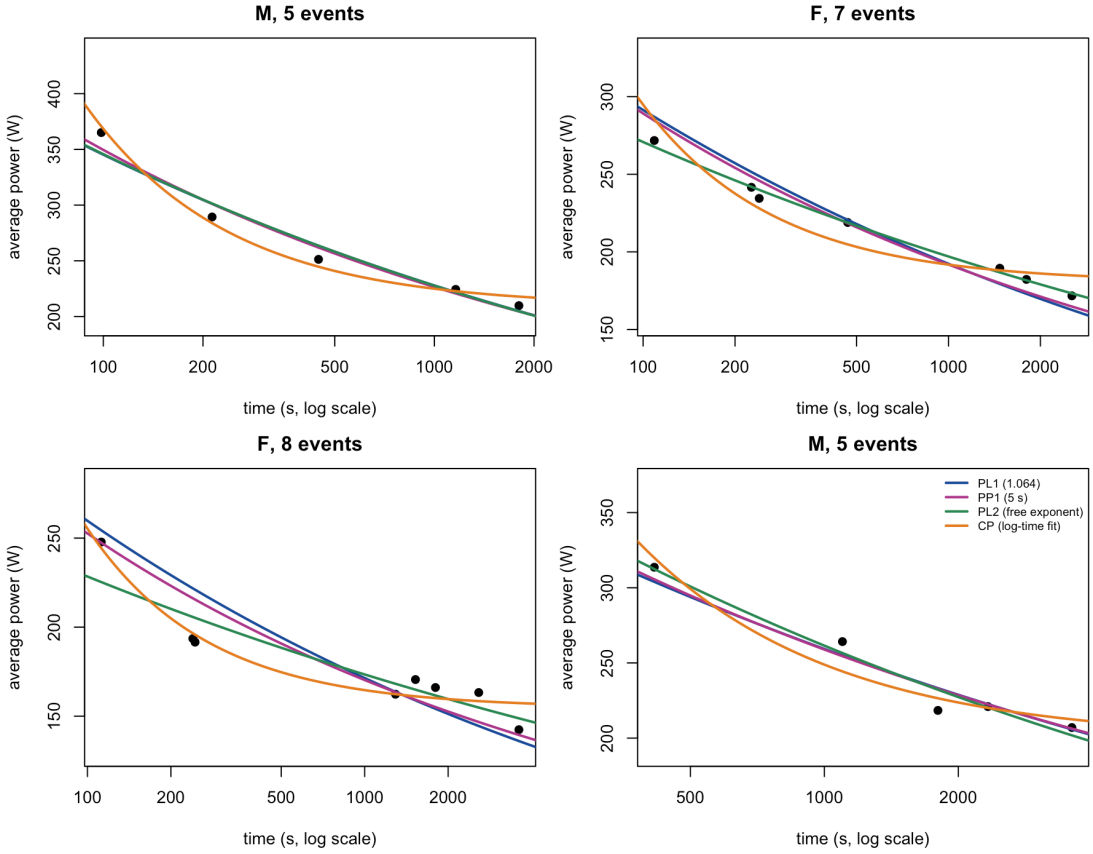


Figure 2: Four randomly chosen athlete-seasons: measured average power against duration with the PL1, PP1, PL2 and CP (log-time fit) curves fitted to all of the athlete's events.

5.2 Like-for-like comparisons

Models should be compared at equal complexity. Table 5 pairs the Paul's Law version of each model with its power-law counterpart, and the best two-parameter models with each other.

Table 5: Paired comparisons of median absolute error (percentage points; 95% bootstrap interval; negative: the first model is more accurate; "a closer": share of predictions where a beats b).

Comparison (a vs b)	n	Error a vs b	Difference a − b	a closer
Paul, fixed constant vs power law, fixed exponent (one parameter each)	17,230	2.19% vs 2.25%	−0.07 [−0.10, −0.04]	53%
Paul vs power law, both calibrated by sex and age group (one parameter each)	17,230	2.22% vs 2.23%	0.00 [−0.03, 0.02]	50%
Paul vs power law, both free per athlete (two parameters each)	17,230	2.14% vs 2.20%	−0.05 [−0.08, −0.03]	55%
Paul, free constant vs critical power (two parameters each)	17,230	2.14% vs 3.02%	−0.88 [−0.94, −0.80]	63%
Critical power vs power law, free exponent (two parameters each)	17,230	3.02% vs 2.20%	0.82 [0.75, 0.90]	39%
Paul vs power law from the 2 km result alone (one parameter each)	9,803	2.29% vs 2.47%	−0.18 [−0.22, −0.14]	58%

5.3 The constant in Paul's Law

The published constant of 5 s per 500 m need not be optimal. We estimate it directly, as the pooled within-athlete slope of pace on $\log_2$ distance, by group (Table 6).

Table 6: Best-fitting Paul constant by group, estimated on the data (published value: 5).

Group	Athletes	Constant (s per 500 m per doubling)	95% CI
All athletes	3,993	5.48	[5.38, 5.57]
Women	695	5.73	[5.42, 6.05]
Men	3,298	5.43	[5.34, 5.52]
≤18	99	6.51	[5.97, 7.07]
19–29	181	6.53	[6.16, 6.91]
30–39	638	6.11	[5.88, 6.32]
40–49	1,074	5.46	[5.29, 5.63]
50–59	1,039	5.18	[5.01, 5.35]
60+	962	5.26	[5.07, 5.45]
Women 19–29	30	7.42	[6.06, 8.74]
Women 30–39	95	6.49	[5.87, 7.07]
Women 40–49	184	5.55	[5.05, 6.04]
Women 50–59	163	5.38	[4.55, 6.14]
Women 60+	205	5.61	[5.06, 6.22]
Men ≤18	81	6.49	[5.91, 7.04]
Men 19–29	151	6.36	[6.00, 6.72]
Men 30–39	543	6.04	[5.81, 6.27]
Men 40–49	890	5.44	[5.25, 5.62]
Men 50–59	876	5.15	[4.98, 5.31]
Men 60+	757	5.16	[4.97, 5.36]

A decisive question is whether the constant is the same for everyone, as Paul's Law assumes, or grows with pace, as a power law implies. Athletes were binned by their 2 km-equivalent pace and the constant estimated within each bin (Table 7, Figure 3).

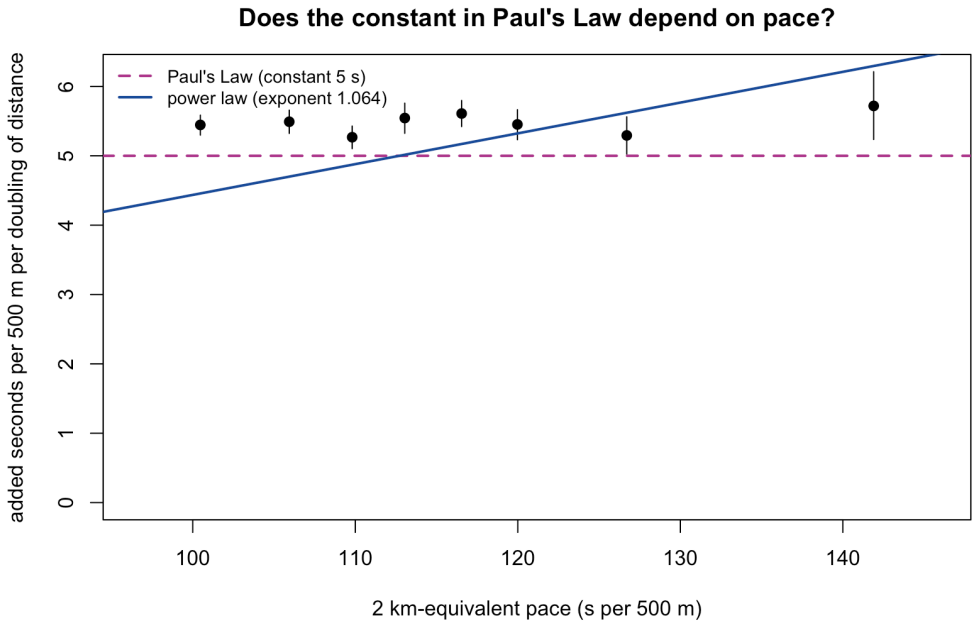


Figure 3: Added seconds per 500 m per doubling of distance, by the athletes' 2 km-equivalent pace, with 95% bootstrap intervals. Dashed: Paul's Law (constant). Solid: power law with exponent 1.064 (proportional to pace).

Table 7: Paul constant by pace bin, and the power-law exponent that corresponds to it at the bin's median pace.

Pace bin (s/500 m)	Athletes	Median pace	Constant (95% CI)	Equivalent exponent
[89.2, 104]	500	100.5	5.45 [5.30, 5.59]	1.078
(104, 108]	499	105.9	5.49 [5.32, 5.66]	1.075
(108, 111]	499	109.8	5.27 [5.10, 5.43]	1.069
(111, 115]	499	113.0	5.54 [5.32, 5.76]	1.071
(115, 118]	499	116.5	5.61 [5.42, 5.80]	1.069
(118, 123]	499	120.0	5.45 [5.23, 5.66]	1.066
(123, 132]	499	126.7	5.29 [5.02, 5.56]	1.060
(132, 303]	499	141.9	5.72 [5.24, 6.21]	1.058

Fitting the 8 bins with a single constant gives 5.45 s and a weighted sum of squares of 11 fitting a constant proportional to pace (a power law with exponent 1.070) gives 149. The constant shape fits better, so the added time per doubling does not grow with pace and Paul's Law describes these data better than a power law. The bins are few and the weights come from bootstrap intervals, so this is a descriptive comparison, not a formal test.

5.4 From the 2 km result alone

Paul's Law was designed to predict from one known effort. Table 8 compares the one-effort versions with the versions that use all the athlete's other efforts.

Table 8: Predictions where the athlete has a 2 km result among the training efforts (9,803 predictions). "From 2 km only" uses just that result; the other models use all other efforts.

Model	Median error	Bias	Difference vs PL1	Wins vs PL1
PP1 (Paul, 5 s per 500 m)	2.11%	0.20%	−0.06 [−0.10, −0.02]	53%
PL1 (calibrated exponent)	2.14%	0.39%	−0.03 [−0.07, 0.01]	54%
PP (calibrated constant)	2.16%	0.40%	−0.02 [−0.06, 0.03]	52%
PL1 (exponent 1.064)	2.17%	0.28%	–	–
PP (from 2 km only)	2.29%	0.39%	0.11 [0.05, 0.19]	47%
PL1 (from 2 km only)	2.47%	0.60%	0.30 [0.23, 0.38]	44%

Using only the 2 km result as the source, Table 9 lists the error for each target event with the published constants of both laws.

Table 9: Median absolute error and median signed error when the 2 km result alone predicts each other event (published constants; positive bias: predicted too slow).

Target event	n	PL1 error	PL1 bias	Paul error	Paul bias
500 m	1,339	4.58%	3.59%	4.21%	3.35%
1000 m	1,508	2.09%	0.10%	2.00%	0.19%
4 min	405	2.20%	−1.24%	2.12%	−1.04%
5000 m	1,921	1.81%	0.81%	1.69%	0.60%
6000 m	845	2.00%	0.09%	1.91%	−0.19%
30 min	1,444	2.48%	−0.03%	2.26%	−0.34%
10,000 m	1,528	2.71%	0.86%	2.46%	0.48%
60 min	813	3.20%	0.09%	3.13%	−0.65%

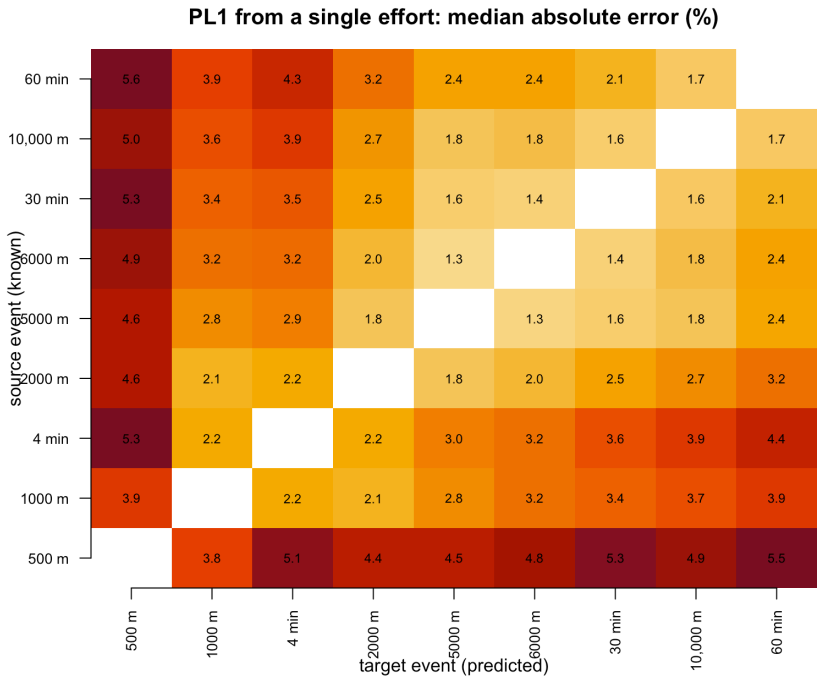


Figure 4: PL1 (exponent 1.064) predicting one event from another single event: median absolute error (%).

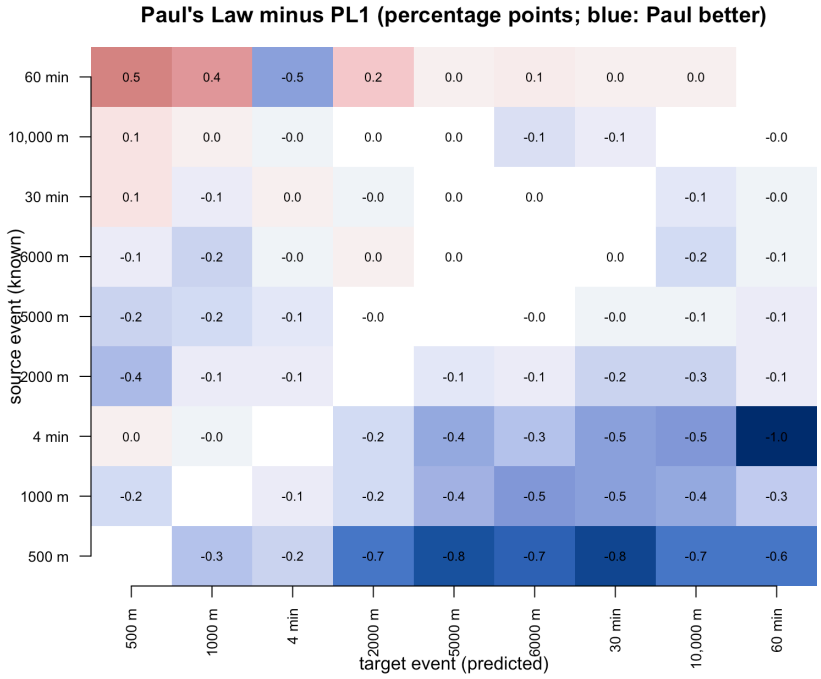


Figure 5: Paul's Law minus PL1, same sources and targets (percentage points of median absolute error; blue: Paul's Law more accurate).

5.5 Interpolation and extrapolation

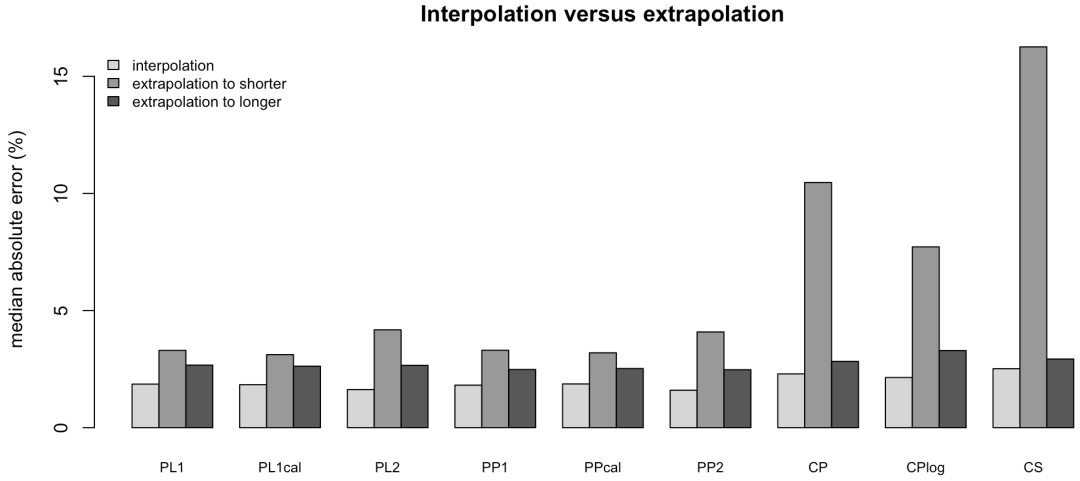


Figure 6: Median absolute error by whether the held-out effort lies inside or outside the range of the training efforts.

Table 10: Median absolute error (%) by position of the held-out effort relative to the training efforts. Bold: lowest in the row.

Held-out effort	n	PL1	PL1cal	PL2	PP1	PPcal	PP2	CP	CP log	CS
Interpolation	9,462	1.86	1.84	1.63	1.82	1.87	**1.60**	2.30	2.14	2.52
Extrapolation to shorter	3,775	3.30	**3.12**	4.18	3.31	3.20	4.09	10.47	7.72	16.26
Extrapolation to longer	3,993	2.67	2.63	2.66	2.49	2.53	**2.48**	2.83	3.29	2.93

Table 11: Median absolute error (%) by spread of the training efforts' durations. Bold: lowest in the row.

Longest / shortest training effort	n	PL1	PL1cal	PL2	PP1	PPcal	PP2	CP	CP log	CS
< 2	576	2.72	**2.62**	8.87	2.76	2.68	9.98	10.48	10.50	11.80
2–4	2,638	2.58	2.50	3.27	**2.48**	2.53	3.29	4.89	4.91	5.61
4–10	4,343	2.28	2.28	**2.24**	2.28	2.25	2.26	3.12	3.05	3.39
≥ 10	9,673	2.15	2.11	1.86	2.07	2.13	**1.80**	2.68	2.59	2.97

Table 12: Median absolute error (%) by number of training efforts. Bold: lowest in the row.

Training efforts	n	PL1	PL1cal	PL2	PP1	PPcal	PP2	CP	CP log	CS
2	4,467	2.56	2.50	3.09	2.51	**2.47**	2.97	4.04	4.05	4.53
3	4,054	2.45	2.37	2.41	2.38	2.37	**2.36**	3.23	3.35	3.56
4+	8,709	2.06	2.03	1.80	1.98	2.07	**1.74**	2.71	2.51	2.97

5.6 By held-out event

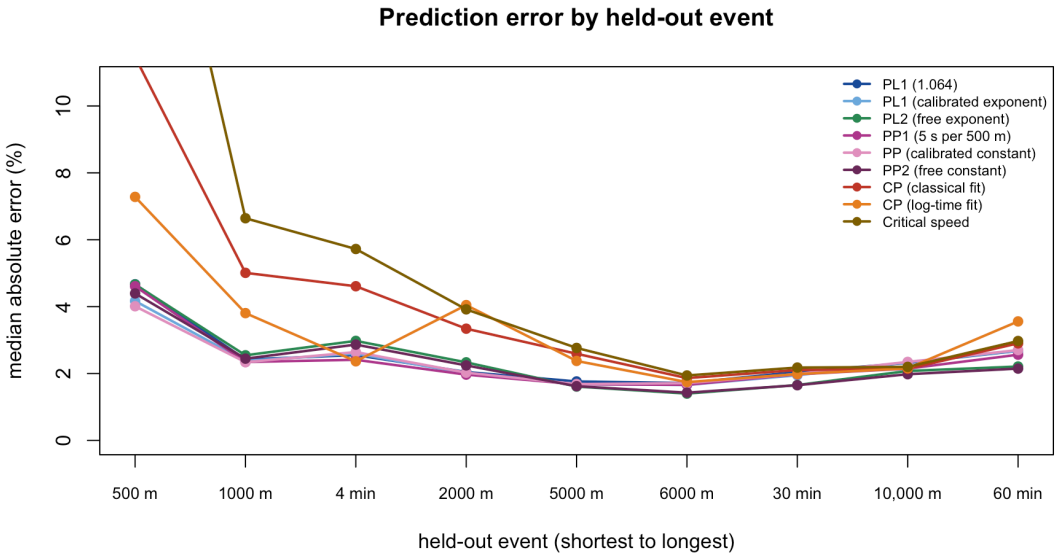


Figure 7: Median absolute error for each held-out event, ordered from shortest to longest typical duration.

Table 13: Median absolute error (%) by held-out event. Bold: lowest in the row.

Held-out event	n	PL1	PL1cal	PL2	PP1	PPcal	PP2	CP	CP log	CS
500 m	1,877	4.67	4.17	4.66	4.61	**4.01**	4.39	11.46	7.28	20.92
1000 m	2,337	2.42	2.38	2.54	2.34	**2.34**	2.44	5.01	3.81	6.64
4 min	659	2.55	2.60	2.98	2.41	2.64	2.86	4.61	**2.36**	5.72
2000 m	2,769	2.04	2.00	2.33	**1.97**	2.02	2.24	3.34	4.05	3.92
5000 m	2,549	1.76	1.68	**1.61**	1.68	1.68	1.62	2.59	2.38	2.76
6000 m	1,295	1.72	1.66	**1.40**	1.66	1.71	1.43	1.86	1.74	1.94
30 min	2,190	2.05	1.95	1.65	1.98	1.98	**1.65**	2.10	1.98	2.18
10,000 m	2,260	2.32	2.34	2.08	2.15	2.35	**1.98**	2.14	2.16	2.20
60 min	1,294	2.68	2.68	2.21	2.56	2.70	**2.15**	2.90	3.56	2.97

The lowest-error model differs by event: 500 m: PPcal, 1000 m: PPcal, 4 min: CP log, 2000 m: PP1, 5000 m: PL2, 6000 m: PL2, 30 min: PP2, 10,000 m: PP2, 60 min: PP2.

5.7 Critical power on its usual range

The CP model is usually applied to efforts of about 2 to 20 minutes. Table 14 adds the variant fitted only to such efforts.

Table 14: The main models and CP restricted to 2–20 minute training efforts, on the 8,901 predictions where all are defined.

Model	Median error	Bias	Difference vs PL1	Wins vs PL1
PP2 (free constant)	1.92%	0.22%	−0.28 [−0.35, −0.21]	54%
PL2 (free exponent)	1.99%	0.24%	−0.21 [−0.29, −0.14]	53%
PP1 (Paul, 5 s per 500 m)	2.13%	0.22%	−0.07 [−0.11, −0.03]	54%
PP (calibrated constant)	2.19%	0.42%	−0.01 [−0.06, 0.03]	52%
PL1 (calibrated exponent)	2.19%	0.38%	−0.01 [−0.05, 0.03]	53%
PL1 (exponent 1.064)	2.20%	0.32%	–	–
CP (log-time fit)	2.64%	−0.53%	0.45 [0.34, 0.54]	44%
CP (classical fit)	2.79%	−0.64%	0.59 [0.48, 0.69]	40%
Critical speed	3.08%	−0.73%	0.88 [0.76, 1.01]	37%
CP (2–20 min efforts only)	3.61%	−2.30%	1.42 [1.29, 1.57]	34%

On this set CP restricted to 2–20 minute efforts is less accurate than PL1 (3.61% against 2.20%; difference 1.42 percentage points [1.29, 1.57]).

5.8 Fit in the athlete's own data and fitted parameters

Table 15 asks how well each model describes an athlete's efforts in the data it was fitted to, adjusted for the number of parameters (residual standard deviation of log time with $n - p$ degrees of freedom).

Table 15: In-sample residual scatter and, among the two-parameter models, the share of athlete-seasons in which each fits best.

Model	Parameters	Median residual SD	Best two-parameter fit
PL1 (exponent 1.064)	1	2.74%	–
PL2 (free exponent)	2	1.89%	43%
PP1 (Paul, 5 s per 500 m)	1	2.64%	–
PP2 (free constant)	2	1.82%	26%
CP (classical fit)	2	4.26%	–
CP (log-time fit)	2	2.41%	28%
Critical speed	2	6.09%	3%

Fitting each athlete-season to all of its events, the median exponent of the free power law is 1.067 (interquartile range 1.050–1.085); the median free Paul constant is 5.48 s per 500 m per doubling (interquartile range 4.18–6.89). The median critical power is 170 W and the median W′ is 19.3 kJ (interquartile range 12.9–28.1 kJ); 2.1% of classical CP fits had CP or W′ at the zero boundary.

5.9 Sex and age

Table 16: Median absolute error (%) by sex. Bold: lowest in the row.

Sex	n	PL1	PL1cal	PL2	PP1	PPcal	PP2	CP	CP log	CS
Women	3,029	2.41	2.37	2.43	**2.32**	2.37	2.41	3.57	3.35	3.89
Men	14,201	2.21	2.19	2.15	2.16	2.19	**2.09**	3.06	2.96	3.43

Table 17: Median absolute error (%) by age group. Bold: lowest in the row.

Age group	n	PL1	PL1cal	PL2	PP1	PPcal	PP2	CP	CP log	CS
≤18	372	2.57	**2.54**	2.82	2.68	2.74	2.75	4.81	4.80	5.19
19–29	658	2.35	**2.12**	2.42	2.21	2.17	2.52	4.39	4.46	4.87
30–39	2,508	2.57	2.51	2.62	2.50	**2.49**	2.54	3.63	3.67	4.18
40–49	4,602	2.33	2.34	2.30	2.29	2.30	**2.23**	3.29	3.14	3.66
50–59	4,556	2.18	2.19	2.11	2.13	2.16	**2.06**	3.09	2.95	3.38
60+	4,534	2.10	2.00	1.93	2.01	2.07	**1.86**	2.60	2.42	2.84

6 Discussion

Paul's Law and the power law. With the published constants Paul's Law predicts held-out efforts better than the power law with exponent 1.064. After calibrating both on other athletes, the two are not clearly different. When each athlete gets a free exponent or constant, Paul's Law is again more accurate. The laws are close over a limited range of distances and athlete paces (Table 1), so small differences are to be expected; they differ most when the athlete's pace is far from the pace at which Paul's constant happens to match the power law's exponent.

Is the constant constant? The data on the constant (Table 7) address the one assumption on which the two laws differ: added time per doubling stays roughly fixed across paces, as Paul's Law assumes.

Calibration. The constants of both laws were treated as tunable; calibrated by sex and age group on other athletes they are free to adapt to the population. The comparison of the calibrated and uncalibrated versions in Table 4 shows how much of the error of each published law is due to its constant rather than its form.

What a second parameter buys. The critical-power model is less accurate than the one-parameter power law. CP implies a power that levels off at a critical value for long efforts and rises without bound for short ones, while the power laws decline smoothly over all durations. The CP model is normally applied to efforts of about 2 to 20 minutes, whereas the events here run from about 1.5 to 60 minutes; the restricted fit in Table 14 tests that.

Interpolation and extrapolation. A two-parameter model has more freedom and, with two or three training efforts, little data to pin it down. Table 10 and Table 11 show where that matters. Predictions that extrapolate

beyond the training range are the usual practical need (for example a 10 km time from a 2 km result) and are where the choice of functional form matters most.

Effort. All models assume maximal efforts. A sub-maximal result can distort a fit to two or three efforts, either by being absorbed into a distorted curve or by being amplified in extrapolation. The analysis cannot separate this from true model error; the errors reported include both.

Limitations. The athletes are those with at least three verified events across a threefold range of distance, mostly male masters rowers, from one season. The comparison is on the typical error of one held-out result, not the best case for any athlete. The calibrated constants are cross-fitted by athlete but come from the same data source as the evaluation. The pace-bin analysis uses eight bins and bootstrap weights, and pace is the athlete's 2 km-equivalent pace under a fixed exponent. Intervals reflect sampling variation only.

7 Conclusion

On 17,230 held-out efforts the model with the lowest median error is PP2 (free constant) (2.14%); the one-parameter power law has 2.25% and Paul's Law with the published constant 2.19%. Paul's Law is, to first order, a power law whose exponent depends on pace (Table 1). In these data the added time per doubling stays close to fixed across paces, which favours Paul's constant over the power law's proportional form. A critical-power model is less accurate than the one-parameter laws on these data. The one-parameter models need only a single effort, which counts in their favour when a second well-spaced maximal effort is not available. The comparison does not say that any model is physiologically correct, only how well each predicts the next effort.

8 Appendix: reproducibility

Run `power_law_analysis.R` first (it prepares the data, defines the athlete-season eligibility rule and writes the results that Table 2 uses), then `cp_vs_powerlaw.R`, which sources it and writes `results/cp_*.csv`, `results/pp_*.csv`, `results/cp_key_numbers.json` and `results/fig_cp_*.png`, `results/fig_pp_*.png`. Compile this report with `typst compile report_cp.typ`. Settings: 1,000 bootstrap resamples, 10-fold cross-fitting of the calibrated constants, 8 pace bins, Paul's constant 5 s per 500 m, power constant 2.8, CP efforts restricted to 2–20 minutes in the restricted variant. The predictions file contains an anonymous athlete index rather than athlete identifiers.